RASOOL SAGHALEYNI, PhD

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I hold EU (Swedish) citizenship, and I am based in Switzerland (Visp) in case that it matters for hiring requirements.

PROFESSIONAL SUMMARY

PhD-trained data scientist and biotechnologist with strong experience in advanced analytics, **statistical modeling**, reproducible **workflows**, and data visualization for complex biological datasets. Experienced in developing **Python**- and **R**-based analytical pipelines, **dashboards**, and structured reports that help teams understand variability, identify patterns, and support data-driven decisions. I am particularly motivated to apply process analytics, multivariate analysis, trend monitoring, and predictive modeling to support MSAT, Process Development, and Operations in improving process understanding and operational performance.

CORE COMPETENCIES

- **Process Analytics & Operational Data Science** – Development of statistical and data-driven analyses to understand process behavior, trends, and performance drivers across biological datasets.
- **Multivariate & Statistical Modeling** – Practical experience with high-dimensional data analysis, dimensionality reduction, clustering, regression, machine learning, trend analysis, and model interpretation using Python and R.
- **Dashboards & Analytical Views** – Experience building interactive dashboards, reports, and analytical tools using R Shiny, Quarto, and reproducible reporting frameworks to support monitoring, interpretation, and technical discussions.
- **Reproducible Data Workflows** – Advanced experience developing scalable and well-documented workflows using Python, R, Bash, Nextflow, Git, Docker/Apptainer, HPC, and cloud environments.
- **Cross-Functional Communication** – Proven ability to work with experimental scientists, technical teams, infrastructure teams, and external stakeholders.

EMPLOYMENT HISTORY

Staff Scientist - SciLifeLab | *Sep 2022 – Present, Sweden*

- Developed reproducible analytical workflows in Python and R to analyze complex biological and process-related datasets.
- Built dashboards, structured reports, and analytical views using R Shiny, Quarto, and reproducible reporting tools to help stakeholders monitor results, identify trends, and communicate findings clearly.
- Applied statistical modeling, multivariate analysis, dimensionality reduction, clustering, and machine learning methods to identify patterns, variability, and meaningful drivers in high-dimensional datasets.
- Supported troubleshooting and root-cause-oriented discussions by analyzing data trends, identifying anomalies, and translating complex analytical results into clear technical recommendations.
- Worked as a central computational contact across multiple parallel projects, coordinating requirements, priorities, and deliverables with experimental scientists, platform teams, and external collaborators.
- Designed well-documented and version-controlled workflows using Python, R, Git, Nextflow, Snakemake, Docker/Apptainer, HPC, and cloud environments to ensure reproducibility, traceability, and long-term usability.

PhD & Postdoc – Chalmers University in collaboration with AstraZeneca | *Aug 2016 – Aug 2022*

- Applied multi-omics analysis, statistical modeling, and machine learning to study mammalian recombinant protein production in HEK293 and CHO systems.
- Investigated process- and cell-related bottlenecks affecting protein secretion, productivity, and cellular performance, generating insights relevant to mammalian bioprocess development.
- Analyzed high-dimensional experimental datasets to connect process conditions, cell physiology, and productivity-related outcomes.
- Collaborated with experimental and industrial partners to translate analytical findings into biological and process understanding.
- Worked in industry–academia collaboration, contributing to structured reporting and project delivery
- Applied multi-omics analysis to study cellular mechanisms affecting recombinant protein production

Biotechnologist – Analytical Method Development Expert – AryoGen Pharmed | *Jan 2016 – July 2016*

- Supported analytical method development and validation for monoclonal antibody manufacturing, including chromatography-based assays (HPLC/UPLC)
- Contributed to development of bioassays for antibody binding potency (e.g., ELISA-based methods)
- Prepared QC protocols, validation reports, and tech-transfer documentation in GMP-regulated environments
- Participated in deviation investigations, root cause analysis, and audit preparation in collaboration

Junior Biotechnologist – IBRF | *Dec 2014 – Dec 2015*

- Contributed to GMP cleanroom USP operations for biologics production (grades D/C)
- Executed chromatography, virus filtration, and filtration workflows
- Contributed to SOP and recipe development procedures with GDocP compliance

EDUCATION

- **Ph.D.,** Systems Biology Gothenburg, Sweden — Sep 2016 - Sep 2021
- **Master** of Science, Biotechnology, NIGEB, Iran — Aug 2011 - July 2014
- **Bachelor** of Science, Cellular and Molecular Biology, Iran — Aug 2006 - Jun 2010

SELECTED ACHIEVEMENTS

- Developed reproducible monitoring and analysis workflows used to support troubleshooting and data-driven discussions.

SELECTED PUBLICATIONS (Full list here)

- **Cell Culture Engineering** XVIII (2024): Rockberg, J., **Saghaleyni**, R., et al. Cellular demands of secreted protein products–systems and synthetic biology improve quality and titer
- **Cell Reports (2023)**: **Saghaleyni**, R., et al.. Enhanced metabolism and negative regulation of ER stress support higher erythropoietin production in HEK293 cells.

LANGUAGES

English – Work proficiency **| Swedish –** Intermediate **| German –** Beginner (A1.2 learning)